

Yuheng Li

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Education

University of California, San Diego

Master of Science in Computer Science

La Jolla, CA

Expected Dec. 2026

University of California, Los Angeles

Bachelor of Science in Mathematics of Computation

Los Angeles, CA

Jun. 2025

Skills

Programming Languages: Python, C/C++, Java, SQL, MATLAB, Bash

Frameworks & Tools: PyTorch, Hugging Face, LangChain, OpenAI Agents SDK, Git, FastAPI, Docker, AWS

AI/ML: NLP, LLMs, RAG, AI Agents, Post-Training, Machine Learning Systems

Experience

Molex — Machine Learning Engineer Intern

Fremont, CA — Jun. 2026 – Sep. 2026

- Developed a domain-specific **AI Agent** using the **Agents SDK**, **MCP**, and **RAG**, combining live Zemax simulation state with a knowledge base reference to translate natural-language requests into multi-step optical design workflows.
- Engineered LLM **tool-calling** orchestration over a live simulation environment, enabling the agent to execute closed-loop optical design workflows spanning system inspection, experimentation, and performance analysis.
- Designed an agent evaluation framework around an engineer-approved seed set, using **LLMs** to generate additional test cases and the simulator itself to derive **ground truth** for scalable, deterministic evaluation of end-to-end task.

Advance.AI — Machine Learning Engineer Intern

Singapore — Jun. 2025 – Sep. 2025

- Designed an **LLM-driven** data synthesis pipeline using prompt engineering to convert OCR outputs into structured training data for ID documents, scaling labeled samples to **50K+** while reducing annotation time by **~95%**.
- Developed **layout-agnostic** post-processing quality filters to validate LLM outputs and curate the synthetic dataset, achieving **~95%** label accuracy across **~50%** more document layouts with no additional verification effort.
- Fine-tuned a **multimodal** LayoutLM in **PyTorch** on the synthetic dataset by fusing 2D positional embeddings of each token's spatial coordinates with textual signals, increasing field-level **F1-score** by **11%**.

Goldstate Securities Co., Ltd. — Data Scientist Intern

Shenzhen, China — Jul. 2024 – Sep. 2024

- Collaborated with traders to develop data processing pipelines and visualization tools for equity market analysis, enabling efficient exploration of market trends, factor signals, and trading opportunities.
- Built an **LLM-driven** portfolio analysis and daily strategy reporting system.

Projects

Schema Linking for Text-to-SQL | SFT, LoRA, BM25, Hugging Face

Apr. 2026 – Jun. 2026

- Designed a schema-linking pipeline combining LLM-generated **data augmentation** and **BM25** retrieval, dynamically selecting relevant tables under a fixed token budget while achieving **96%** column recall.
- Fine-tuned **Qwen3-1.7B** using **LoRA** on augmented schema-linking data, achieving **0.77 F1** and performance comparable to **DeepSeek V4 Flash** (284B).

RAG over Technical Documentation | RAG, FAISS, Reranking, LangChain

Sep. 2025 – Dec. 2025

- Developed a **RAG** system that combines lexical and semantic retrieval with reranking, improving evidence retrieval quality and achieving approximately **0.87 Precision@5** and **0.81 Recall@5**.
- Built an **LLM-as-a-judge** evaluation pipeline to measure correctness, faithfulness, and completeness, helping identify retrieval failures separately from generation errors.

Research

Decode-Free Information Extraction [GitHub]

Jan. 2026 – Mar. 2026

- Proposed a single-pass information extraction method that classifies passage relevance from **LLM hidden states** at EOS positions, bypassing autoregressive decoding; evaluated on **HotpotQA**.
- Applied **LoRA** fine-tuning on Qwen3-0.6B and Qwen3-Embedding-0.6B backbones, achieving **0.89 F1** with a **14%** absolute improvement over the finetuned encoder baseline on multi-hop evidence selection.

SourceBench | Prof. Yiyang Zhang [arXiv]

Oct. 2025 – Jan. 2026

- Co-authored *SourceBench: Can AI Answers Reference Quality Web Sources?* A benchmark evaluating citation quality and source attribution in generative AI responses.
- Designed an 8-metric evaluation framework covering content quality and page-level signals; calibrated a Qwen-based **LLM-as-a-judge** against human expert labels to enable scalable assessment.